ECON5130 Group Assignment — Group 11

Empirical Asset Pricing via Machine Learning: A Replication Study

Introduction

This study replicates the analysis of Gu, Kelly, and Xiu (2020), which examines whether machine learning methods improve the measurement and prediction of equity risk premiums relative to traditional models. We evaluate several models using a panel of approximately 38,000 U.S. equities from January 1960 to December 2024, sourced from CRSP and the Open Asset Pricing portal. Specifically, we compare regularised linear models, dimension reduction techniques, tree-based methods, and neural networks against a CAPM benchmark in out-of-sample return prediction.

Data and Methodology

Data

The dataset consists of monthly observations for firms listed on NYSE, NASDAQ, and AMEX. Each observation includes stock returns, prices, and 114 firm-level predictors capturing momentum, volatility, liquidity, and valuation. Excess returns are computed using the 3-month U.S. Treasury Bill as the risk-free rate.

Preprocessing

The data cleaning process follows a structured pipeline:

- Remove observations missing identifiers, returns, or dates
- Drop predictors with more than 40% missing values (reducing features from 114 to ~80)
- Impute missing predictors using cross-sectional monthly medians within the same month and year to prevent information leakage across firms;
- Drop remaining incomplete rows
- Rank-normalise predictors to the [-1, 1] range

Rank normalisation is used instead of standard scaling to improve robustness to outliers and ensure comparability across predictors. The CAPM matrix is constructed before normalisation to preserve raw beta values. Due to the lack of a consistent market proxy across the full sample, reported beta values are used directly and left unscaled.

Sample Split

The dataset is divided chronologically:

- Training: January 1960 – December 1999
- Validation: January 2000 – December 2010
- Testing: January 2011 – December 2024

Models are trained on the training set, tuned on validation, and evaluated on testing. CAPM and OLS are estimated only on training data due to the absence of hyperparameters.

Models

We estimate six models:

- **CAPM:** Uses market beta only
- **OLS:** Linear regression with all predictors
- **LASSO:** Applies L1 regularisation with λ tuned over 50 values
- **PCR:** Uses PCA for dimensionality reduction, selecting K (1–20) components
- **Random Forest:** Tuned over depth, leaf size, and feature fraction
- **Neural Networks:** Three architectures (NN1, NN3, NN5) with ReLU, dropout, batch normalisation, Huber loss, early stopping, and ensemble averaging

Evaluation

Performance is measured using out-of-sample R^2 as defined in Gu et al. (2020), benchmarking predictions against zero rather than the historical mean. This avoids artificially inflating performance due to noisy average returns.

We also use the Diebold-Mariano test with Newey-West errors for statistical comparison of forecasting accuracy and report both monthly and annual out-of-sample R^2.

Predictive Performance

Out-of-Sample R² Comparison (Monthly)

Model	Monthly OOS R² (%)
PCR	0.092716
NN3	0.071548
CAPM (benchmark)	0.065019
Lasso	0.052199
RF	-0.140700
OLS	-0.221337

Table 1: Monthly OOS R^2 for All Models

Annual Out-of-Sample R²

Model	Annual OOS R² (%)
NN3	1.3335
Lasso	1.2481
PCR	1.2379
CAPM (benchmark)	1.1944
OLS	-0.6817
RF	-2.2794

Table 2: Annual OOS R^2

Table 1 presents the monthly out-of-sample R^2 for all models. PCR achieves the highest monthly out-of-sample R^2 (0.093%), followed closely by NN3. PCR achieved the highest out-of-sample R^2 at the monthly level, while the neural network performed better at the annual level (See *Table 2*). In contrast, OLS produces a negative R^2 (−0.221%), confirming overfitting in high-dimensional settings.

Random Forest (RF) produces a negative R^2 (-0.866%), suggesting insufficient generalisation. The poor RF performance differs from Gu et al. (2020). When retrained on combined training and validation data, its performance improves significantly. This suggests two possible explanations: a smaller effective dataset compared to Gu et al. (2020), which may cause underfitting and a long gap (11 years) between training and testing periods, which may weaken model stability.

```
Best RF params based on validation OOS R2:
{'n_estimators': 500, 'max_depth': 2, 'min_samples_leaf': 50, 'max_features': 0.1}
Best validation OOS R2: -0.001321
RF Test OOS R2: 0.2048%
```

Despite lower magnitudes overall, the ranking of models aligns with the original paper: regularised and dimension-reduced models outperform OLS, while nonlinear models (neural networks) are competitive or superior.

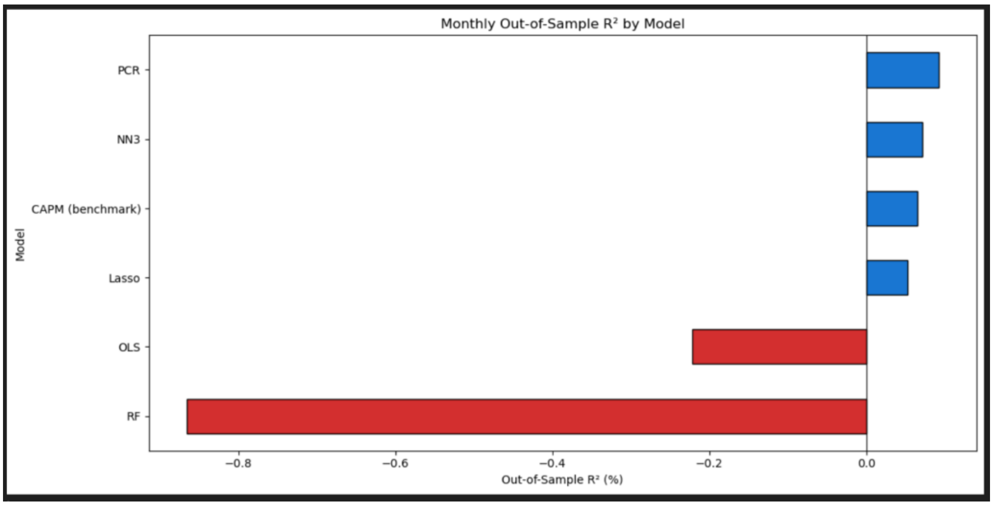

Figure 1: Monthly OOS R^2 Bar Chart

PCR selects $K = 2$ components, explaining 21.3% of total predictor variance. We restricted the component search to $K \leq 20$ to prevent overfitting as higher values ($K > 20$) produce negative validation R^2, indicating capture of noise. This highlights the importance of restricting model complexity in low signal-to-noise environments.

Annual OOS R^2 values in *Table 2* are substantially larger than monthly ones, consistent with the idea that aggregation improves signal relative to noise.

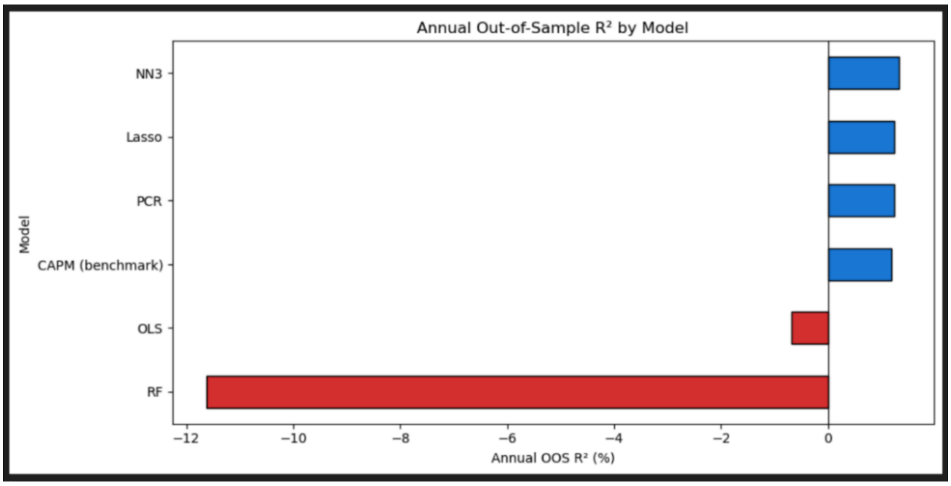

Figure 2: Annual OOS R^2 Bar Chart

Notably, NN5 achieves the highest test R^2 (0.163%), although NN3 is selected based on validation performance. The results table reports NN3 as it was selected by the validation criterion; however, NN5's superior test performance suggests that deeper architectures generalise better in this setting, consistent with the findings of Gu et al. (2020).

Statistical Comparison

Diebold-Mariano Test Statistics
(Positive = column model outperforms row model)
(|t| > 1.96 = significant at 5% level)

	OLS	Lasso	PCR	RF	NN3
OLS	0.00	7.51	7.17	0.50	0.82
Lasso	-7.51	0.00	1.09	-0.65	-0.10
PCR	-7.17	-1.09	0.00	-0.79	-0.23
RF	-0.50	0.65	0.79	0.00	0.21
NN3	-0.82	0.10	0.23	-0.21	0.00

Table 3: Diebold-Mariano Test Statistics

We used the Diebold–Mariano test in order to compare the predictive accuracy of the models. This test evaluates, for each pair of models, whether the difference in their forecasting errors is statistically significant or if it could have occurred by chance. Values greater than 1.96 (or smaller than -1.96) indicate that the difference is statistically significant at the 5% level. (or +-1.645 for 10% significance level)

From the results, we observe that Lasso and PCR both significantly outperform OLS. This confirms that more advanced models provide better predictive performance than the simple linear benchmark. However, the differences between the best performing models are not statistically significant which might mean that their ability to predict is similar.

Variable Importance

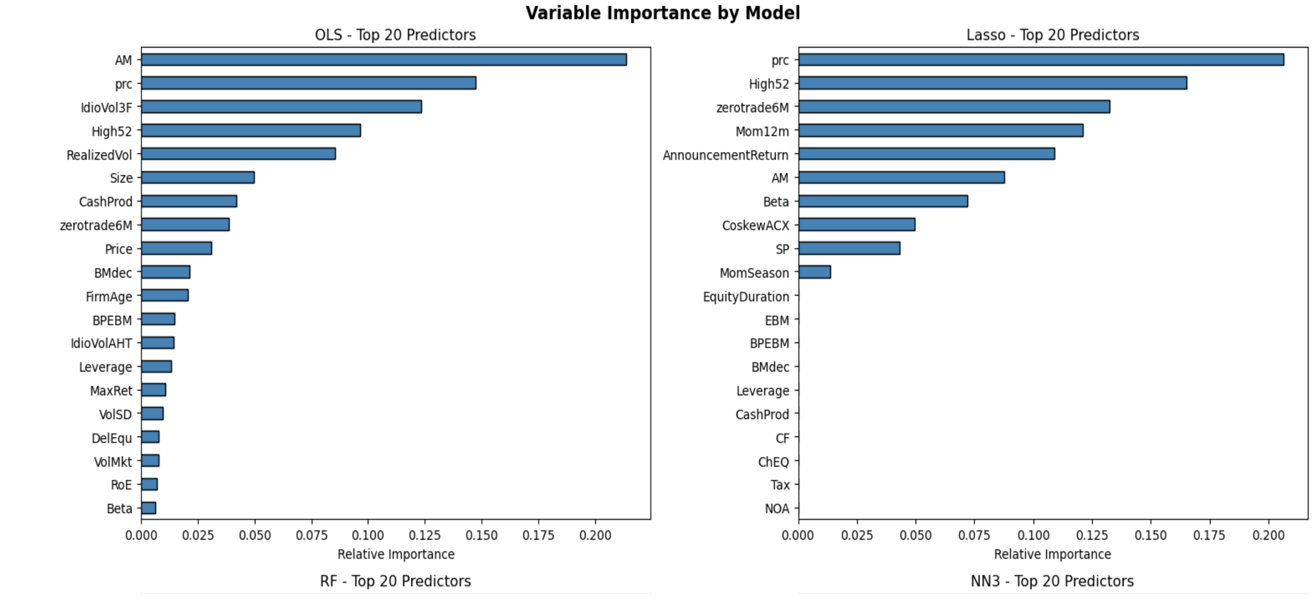

Figure 3: Top 20 Variable Importance (OLS & LASSO)

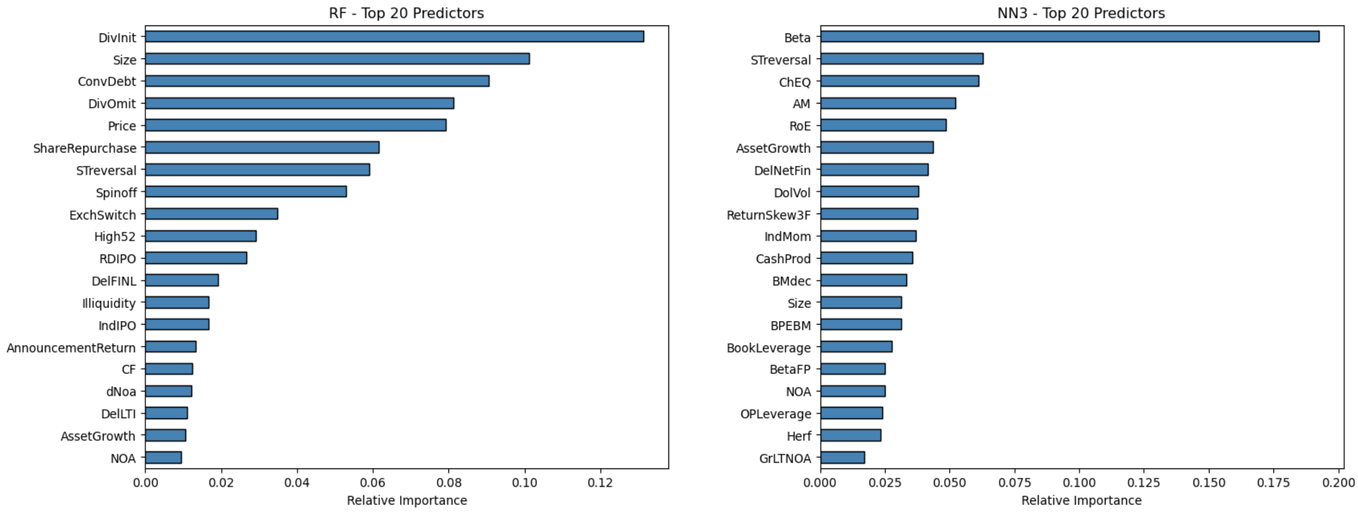

Figure 4: Top 20 Variable Importance (RF & NN3)

Variable importance is assessed using a reduction-in-R^2 approach, setting each predictor to zero and measuring the drop in predictive accuracy, while Random Forest uses mean decrease in impurity. Across models, key predictors fall into four categories consistent with Gu et al. (2020):

- **Price Trends:** momentum and short-term reversal
- **Liquidity:** trading volume, bid-ask spread, zero-trading days
- **Volatility:** realised and idiosyncratic volatility, beta
- **Valuation Ratios:** earnings-to-price, sales-to-price, asset-to-market

Portfolio Analysis

OLS - Decile Portfolio Performance

	Pred	Avg	SD	SR
1	-0.0173	-0.0042	0.0586	-0.2483
2	-0.0063	0.0006	0.0498	0.0411
3	-0.0012	0.0026	0.0454	0.1975
4	0.0026	0.0040	0.0442	0.3119
5	0.0057	0.0039	0.0436	0.3085
6	0.0087	0.0047	0.0457	0.3533
7	0.0118	0.0053	0.0467	0.3958
8	0.0152	0.0068	0.0480	0.4895
9	0.0196	0.0073	0.0503	0.5008
10	0.0289	0.0115	0.0611	0.6541
H-L	0.0462	0.0157	0.0311	1.7550

Table 4: OLS Portfolio Performance

Lasso - Decile Portfolio Performance

	Pred	Avg	SD	SR
1	-0.0073	-0.0049	0.0556	-0.3035
2	-0.0005	-0.0002	0.0473	-0.0181
3	0.0025	0.0030	0.0431	0.2444
4	0.0046	0.0039	0.0412	0.3313
5	0.0062	0.0044	0.0424	0.3579
6	0.0077	0.0049	0.0440	0.3880
7	0.0093	0.0050	0.0468	0.3722
8	0.0111	0.0065	0.0521	0.4298
9	0.0139	0.0073	0.0627	0.4020
10	0.0203	0.0129	0.0654	0.6841
H-L	0.0275	0.0178	0.0351	1.7554

Table 5: LASSO Portfolio Performance

PCR - Decile Portfolio Performance

	Pred	Avg	SD	SR
1	0.0011	0.0003	0.0826	0.0145
2	0.0036	-0.0034	0.0730	-0.1635
3	0.0049	0.0002	0.0674	0.0124
4	0.0059	0.0006	0.0599	0.0324
5	0.0067	0.0048	0.0529	0.3168
6	0.0075	0.0046	0.0451	0.3525
7	0.0081	0.0064	0.0408	0.5459
8	0.0088	0.0070	0.0371	0.6534
9	0.0097	0.0079	0.0337	0.8160
10	0.0114	0.0107	0.0335	1.1086
H-L	0.0103	0.0104	0.0613	0.5869

Table 6: PCR Portfolio Performance

RF - Decile Portfolio Performance

	Pred	Avg	SD	SR
1	0.0063	0.0052	0.0402	0.4489
2	0.0066	0.0063	0.0394	0.5535
3	0.0072	0.0057	0.0383	0.5137
4	0.0082	0.0040	0.0382	0.3627
5	0.0086	0.0035	0.0426	0.2842
6	0.0092	0.0045	0.0468	0.3352
7	0.0103	0.0048	0.0526	0.3195
8	0.0127	0.0046	0.0570	0.2826
9	0.0157	0.0040	0.0566	0.2422
10	0.0207	0.0036	0.0859	0.1470
H-L	0.0143	-0.0016	0.0657	-0.0826

Table 7: RF Portfolio Performance

NN3 - Decile Portfolio Performance

	Pred	Avg	SD	SR
1	-0.0284	-0.0098	0.0845	-0.4013
2	-0.0102	-0.0005	0.0637	-0.0248
3	-0.0022	0.0047	0.0428	0.3842
4	0.0012	0.0037	0.0353	0.3616
...				
8	0.0075	0.0083	0.0454	0.6304
9	0.0101	0.0103	0.0543	0.6558
10	0.0187	0.0163	0.0619	0.9105
H-L	0.0471	0.0260	0.0434	2.0802

Table 8: NN Portfolio Performance

Stocks are sorted into deciles based on predicted returns, and a long-short strategy is constructed by buying the top decile and shorting the bottom.

Following the methodology of Gu et al. (2020), portfolios are value-weighted using market capitalisation to ensure results are not driven by small, illiquid stocks. *Tables 4 - 8* report the predicted return, average realised return, standard deviation, and annualised Sharpe ratio for each decile. A key test of model quality is whether realised returns increase monotonically from decile 1 (lowest predicted) to decile 10 (highest predicted). PCR and neural network models *(Tables 6 and 8)* exhibit the strongest monotonic pattern, whereas OLS and RF *(Tables 4 and 7)* show weaker or inconsistent sorting, reflecting their poorer out-of-sample predictive accuracy.

For the best-performing models, realised returns increase monotonically across deciles. The cumulative returns plot *(Figure 6)* visually demonstrates the divergence between the long and short sides over the test period, with Neural Network and PCR portfolios showing the most consistent performance. Long-short strategies generate positive cumulative returns, with Sharpe ratios indicating economically meaningful gains from machine learning. These results demonstrate that even small predictive improvements in R^2 translate into significant portfolio performance.

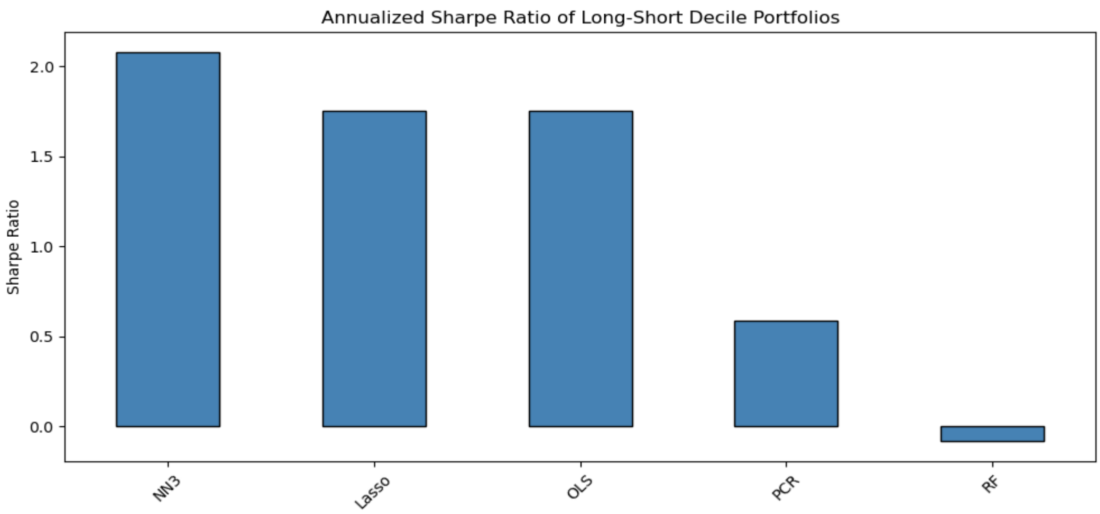

Figure 5: Annualised Sharpe Ratio of Decile Portfolios

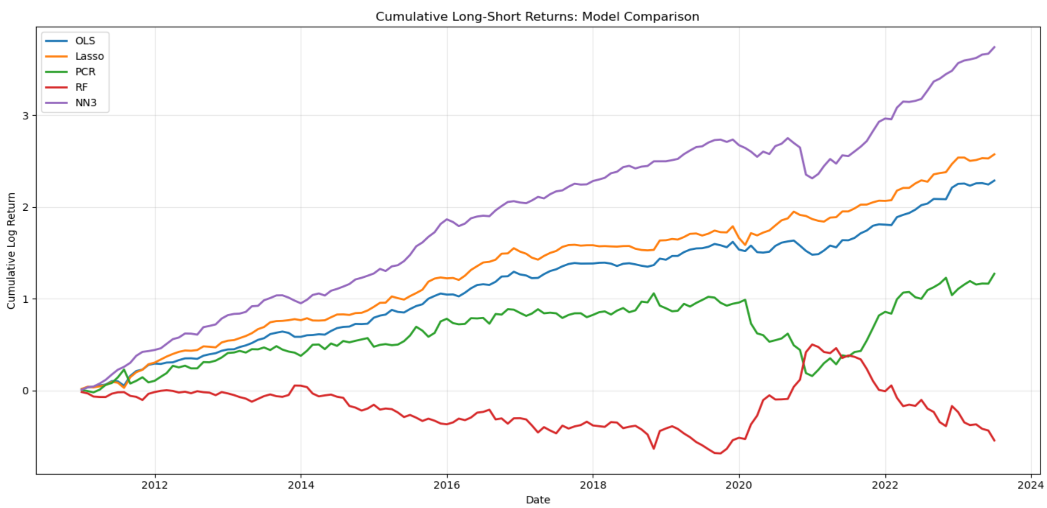

Figure 6: Cumulative Returns across Models

Limitations & Conclusion

Several limitations merit acknowledgement:

- First, we use a fixed train/validation/test split rather than the paper's expanding-window approach, which may reduce the stability of our estimates.
- Second, computational constraints limited our neural network training to 100 epochs with ensembles of three; larger ensembles and more extensive hyperparameter searches could improve NN performance.
- Third, we dropped predictors with greater than 40% missing values, which excluded some economically relevant characteristics.
- Finally, our predictor set does not include the macroeconomic interaction terms used in the paper (94 characteristics × 8 macro variables), which the authors identify as important for capturing time-varying risk premiums.

Despite these constraints, the results confirm the central finding of Gu et al. (2020): machine learning methods, particularly dimension reduction and neural networks, improve equity risk premium prediction relative to traditional models.

Importantly, the dominant predictive signals, namely, momentum, liquidity, and volatility, are consistent across methods, reinforcing their robustness as drivers of expected returns.

References

1. Gu, S., Kelly, B., & Xiu, D. (2020). Empirical Asset Pricing via Machine Learning. *The Review of Financial Studies*, *33*(5), 2223–2273. https://doi.org/10.1093/rfs/hhaa009